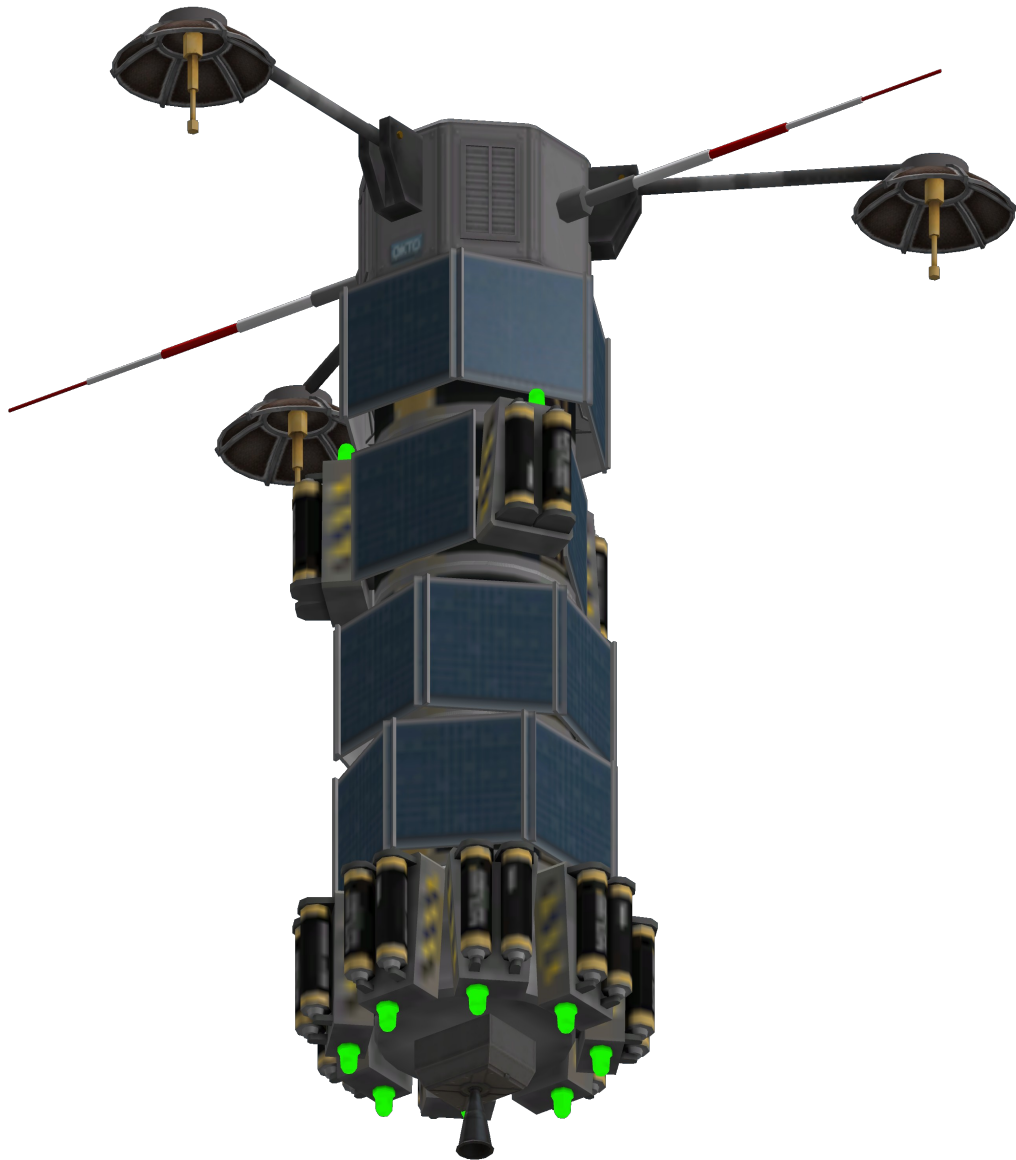


Model 1123 communications satellite

Specifications and assembly instructions



Description

The model 1123 satellite is an HG-5 based relay satellite that takes advantage of the wide angle of the HG-5 antenna to cover a large footprint on planetary surfaces. The presence of a second antenna allows the satellite to relay to off-world targets, to act as a close range (Mun to Kerbin) relay or as part of a larger communications network.

Variants

The satellite variants can be identified by the following suffixes

Model 1123[Y]

Where Y defines the attachment point and can be

Suffix	Type	Notes
(no suffix)	No attachment point	
K	KAS JS-1 socket	KAS only
C	Clamp-o-Tron jr	

Uses

TSC satellite

Mun-Kerbin relay

STC Satellite

General relay

Costs and tech

Costs

Base model	8365	Funds
Model K	9115	Funds
Model C	9165	Funds

Tech requirements

Technology	Use	Notes
Electrics	Probodobodyne OKTO Z-200 Battery OX-STAT Panel	
Basic science	Z-100 Battery HG-5 Antenna	
Engineering 101	Communotron 16	

Propulsion systems	LV-1 Ant engine	
	OSCAR-B tank	
Miniaturization	Clamp-O-Tron jr.	Model C only
Advanced construction	JS-1 socket	Model K only

Specifications

General specifications

Parameter	Value	Units	Notes
Electrical storage	2710	EC	
Electrical storage margin	15	%	kerbostationary orbit
	427	EC	kerbostationary orbit
Max orbital alt (circular)	4300	Km	limited by battery storage
Min orbital alt (circular)	640	Km	limited by battery charger
Power consumption (absolute max)	1.96	EC/sec	
Power consumption (deployment)	0.28	EC/sec	Omnis on, HG5 off, OKTO on, wheels idle
Power consumption (mission)	1.91	EC/sec	OKTO hibernating, wheels off, radios full on
Power generation (max)	2.73	EC/sec	Assumes 0.4 EC for 100% exposure
	6.83	%Exposure	
Power generation (min)	2.61	EC/sec	assumes panel ring oriented towards sun
	6.53	%Exposure	assumes panel ring oriented towards sun
Power generation (avg)	2.67	EC/sec	assumes panel ring oriented towards sun
	6.65	%Exposure	assumes panel ring oriented towards sun
Thrust	2	kN	
dV budget (vacuum)	951	m/s	1122C
	969	m/s	1122K
	981	m/s	1122
Mass (wet)	825	Kg	1122C

Parameter	Value	Units	Notes
	813	Kg	1122K
	805	Kg	1122
Mass (dry)	625	Kg	1122C
	613	Kg	1122K
	605	Kg	1122
Omni Antenna range	1875	Km	assumes root model, 0.5 range multiplier and 0.5 multiple antenna bonus

Power budget

All consumption in EC/second.

System	Unit consumption	Amount	Total consumption	Notes
Communotron 16	0.13	2	0.26	
HG-5	0.55	3	1.65	
Mission subtotal			1.91	
OKTO	0.02	1	0.02	
Systems subtotal			1.93	
Reaction wheel	0.03	1	0.03	
Abs. maximum			1.96	

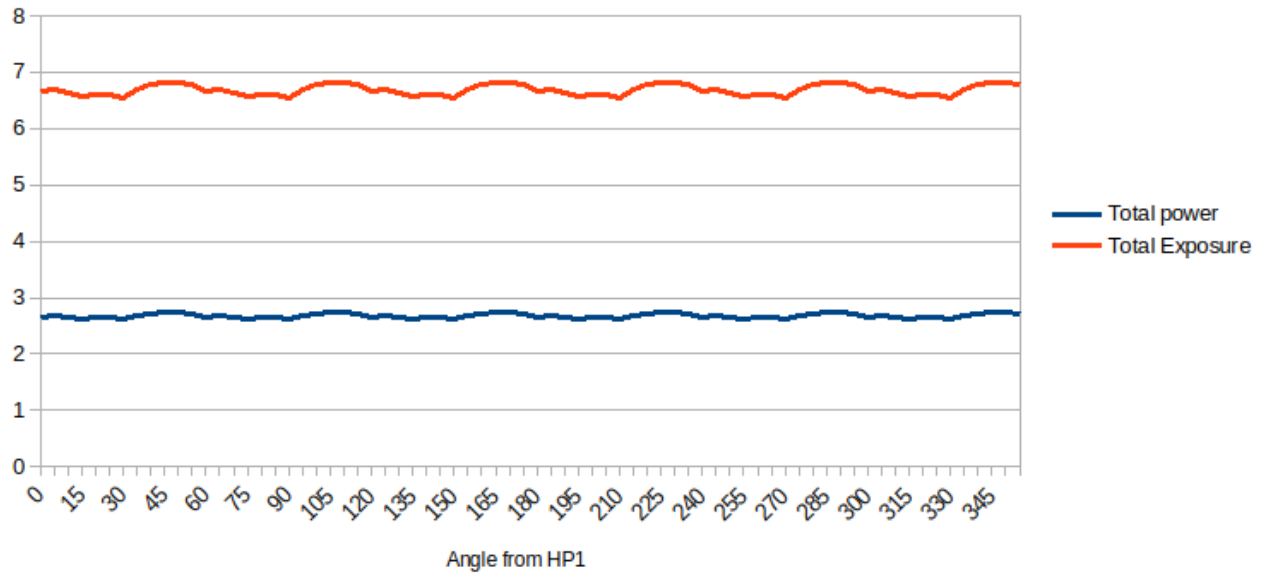
Electrical generator performance

All figures assume side of power rings pointing towards the sun. Derate by cosine of angle to perpendicular of panels in other cases. Mission attitude is satellite perpendicular to ecliptic (direction measured as a line intersecting centre of computer section and centre of propulsion section).

Zero angle is at hardpoint 1 (where the computer label is).

Model 1123 communications satellite

Power generation



Angle	Total power	Total Exposure	UPSR	MPSR	LPSR	T3R
0	2.66	6.66	2	1.93	1.73	1
5	2.67	6.68	1.99	1.88	1.81	1
10	2.66	6.64	1.97	1.81	1.88	0.98
15	2.62	6.56	1.93	1.73	1.93	0.97
20	2.64	6.6	1.88	1.81	1.97	0.94
25	2.64	6.59	1.81	1.88	1.99	0.91
30	2.61	6.53	1.73	1.93	2	0.87
35	2.67	6.68	1.81	1.97	1.99	0.91
40	2.71	6.78	1.88	1.99	1.97	0.94
45	2.73	6.83	1.93	2	1.93	0.97
50	2.73	6.82	1.97	1.99	1.88	0.98
55	2.71	6.77	1.99	1.97	1.81	1
60	2.66	6.66	2	1.93	1.73	1
65	2.67	6.68	1.99	1.88	1.81	1
70	2.66	6.64	1.97	1.81	1.88	0.98
75	2.62	6.56	1.93	1.73	1.93	0.97
80	2.64	6.6	1.88	1.81	1.97	0.94
85	2.64	6.59	1.81	1.88	1.99	0.91
90	2.61	6.53	1.73	1.93	2	0.87
95	2.67	6.68	1.81	1.97	1.99	0.91
100	2.71	6.78	1.88	1.99	1.97	0.94
105	2.73	6.83	1.93	2	1.93	0.97
110	2.73	6.82	1.97	1.99	1.88	0.98
115	2.71	6.77	1.99	1.97	1.81	1
120	2.66	6.66	2	1.93	1.73	1
125	2.67	6.68	1.99	1.88	1.81	1
130	2.66	6.64	1.97	1.81	1.88	0.98
135	2.62	6.56	1.93	1.73	1.93	0.97

140	2.64	6.6	1.88	1.81	1.97	0.94
145	2.64	6.59	1.81	1.88	1.99	0.91
150	2.61	6.53	1.73	1.93	2	0.87
155	2.67	6.68	1.81	1.97	1.99	0.91
160	2.71	6.78	1.88	1.99	1.97	0.94
165	2.73	6.83	1.93	2	1.93	0.97
170	2.73	6.82	1.97	1.99	1.88	0.98
175	2.71	6.77	1.99	1.97	1.81	1
180	2.66	6.66	2	1.93	1.73	1
185	2.67	6.68	1.99	1.88	1.81	1
190	2.66	6.64	1.97	1.81	1.88	0.98
195	2.62	6.56	1.93	1.73	1.93	0.97
200	2.64	6.6	1.88	1.81	1.97	0.94
205	2.64	6.59	1.81	1.88	1.99	0.91
210	2.61	6.53	1.73	1.93	2	0.87
215	2.67	6.68	1.81	1.97	1.99	0.91
220	2.71	6.78	1.88	1.99	1.97	0.94
225	2.73	6.83	1.93	2	1.93	0.97
230	2.73	6.82	1.97	1.99	1.88	0.98
235	2.71	6.77	1.99	1.97	1.81	1
240	2.66	6.66	2	1.93	1.73	1
245	2.67	6.68	1.99	1.88	1.81	1
250	2.66	6.64	1.97	1.81	1.88	0.98
255	2.62	6.56	1.93	1.73	1.93	0.97
260	2.64	6.6	1.88	1.81	1.97	0.94
265	2.64	6.59	1.81	1.88	1.99	0.91
270	2.61	6.53	1.73	1.93	2	0.87
275	2.67	6.68	1.81	1.97	1.99	0.91
280	2.71	6.78	1.88	1.99	1.97	0.94
285	2.73	6.83	1.93	2	1.93	0.97
290	2.73	6.82	1.97	1.99	1.88	0.98
295	2.71	6.77	1.99	1.97	1.81	1
300	2.66	6.66	2	1.93	1.73	1
305	2.67	6.68	1.99	1.88	1.81	1
310	2.66	6.64	1.97	1.81	1.88	0.98
315	2.62	6.56	1.93	1.73	1.93	0.97
320	2.64	6.6	1.88	1.81	1.97	0.94
325	2.64	6.59	1.81	1.88	1.99	0.91
330	2.61	6.53	1.73	1.93	2	0.87
335	2.67	6.68	1.81	1.97	1.99	0.91
340	2.71	6.78	1.88	1.99	1.97	0.94
345	2.73	6.83	1.93	2	1.93	0.97
350	2.73	6.82	1.97	1.99	1.88	0.98
355	2.71	6.77	1.99	1.97	1.81	1

Assembly instructions

The satellite is based on the SpaceBus 1 satellite design and can be built using the standard parts. The following table presents the modules stacked in order, with angle of rotation relative to the computer module.

Module	Type	Angle	Notes
Computer	Computer	0	Module defines zero angle
UPS	Type 1 PSU	0	
T3	Type 3 PSU	0	HG-5 rest on top of OX-STAT
MPS	Type 1 PSU	15	
LPS	Type 1 PSU	30	
Propulsion	Type 1 propulsion	0	

Hardpoint	Device	Notes
1	HG-5 antenna	Use 3x symmetry. Apply on HP 1.
2	Empty	
3	Empty	
4	HG-5 antenna	Not in panel centre. Use 3x symmetry. Apply on HP 1.
5	Empty	
6	HG-5 antenna	

Apply 8x Z-100 batteries on propulsion unit hardpoints, light facing towards exhaust.
 Apple 3x Z-100 batteries on type 3 PSU opposite to panels.